Configuração 10: $M = 256$, $C = 8$, $K = 8$, IoT = 20 dB, 16-QAM

Perda de SNR do BCD-MMSE (iter 1) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas

Fundo laranja no rótulo: o modo com o melhor resultado (estouro: pela varredura de I; quantização: pela varredura de F), comparando de menos bits para mais, sem contar a coluna em que tudo é zero

Faixa "x a y": o outro modo muda o valor (a quantização no mapa de I, o estouro no mapa de F); se os dois extremos têm cores diferentes, a célula é dividida na diagonal (em cima, a cor do menor; embaixo, a do maior)

Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo; sem número onde o formato e o modo não foram sintetizados

TDLA30-5: N = 192, f_c = 3,5 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.46 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.26 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.46 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.82 a 1.83 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.46 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLA30-5: N = 192, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.35 1.9 a 2.1%	0.02 2.0 a 2.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	2.50 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND	×	×	× 1.6 a 1.9%	2.68 1.7 a 2.1%	0.06 2.0 a 2.3%	0.00 2.1 a 2.4%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.67 1.6 a 2.1%	0.06 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.68 1.8 a 2.1%	0.06 2.1 a 2.3%	0.00 2.2 a 2.4%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.66 1.6 a 2.1%	0.06 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.66 1.8 a 2.1%	0.06 2.1 a 2.3%	0.00 2.2 a 2.5%

Bits fracionários F (I = 14)

TDLA30-75: N = 192, f_c = 28 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.44 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.20 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.44 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.08 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.44 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLA30-75: N = 192, f_c = 28 GHz

Quantização	4	6	8	10	12	14
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.60 1.9 a 2.1%	0.03 2.0 a 2.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	2.46 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND	×	×	× 1.6 a 1.9%	2.63 1.7 a 2.1%	0.06 2.0 a 2.3%	0.01 2.1 a 2.4%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.62 1.6 a 2.1%	0.06 1.9 a 2.3%	0.01 2.0 a 2.5%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.64 1.8 a 2.1%	0.06 2.1 a 2.3%	0.01 2.2 a 2.4%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.62 1.6 a 2.1%	0.06 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.62 1.8 a 2.1%	0.06 2.1 a 2.3%	0.00 2.2 a 2.5%

Bits fracionários F (I = 14)

TDLA30-300: N = 168, f_c = 28 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.37 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	2.14 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.37 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	0.98 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.38 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLA30-300: N = 168, f_c = 28 GHz

Quantização	4	6	8	10	12	14
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.48 1.9 a 2.1%	0.03 2.0 a 2.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	2.45 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND	×	×	× 1.6 a 1.9%	2.53 1.7 a 2.1%	0.06 2.0 a 2.3%	0.00 2.1 a 2.4%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.56 1.6 a 2.1%	0.06 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.56 1.8 a 2.1%	0.05 2.1 a 2.3%	0.00 2.2 a 2.4%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.55 1.6 a 2.1%	0.05 1.9 a 2.3%	0.00 2.0 a 2.5%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.55 1.8 a 2.1%	0.05 2.1 a 2.3%	0.00 2.2 a 2.5%

Bits fracionários F (I = 14)

TDLC300-100: N = 132, f_c = 3,5 GHz

Estouro	6	8	10	12
AP_SAT	× 5.5 a 6.6%	× 6.0 a 7.2%	2.09 6.1 a 7.2%	0.00 6.1 a 7.3%
AP_SAT_ZERO	× 6.6%	× 7.1 a 7.2%	1.81 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_SAT_SYM	× 6.5 a 6.6%	× 7.1 a 7.2%	2.09 7.1 a 7.2%	0.00 7.2 a 7.3%
AP_WRAP	× 5.8 a 6.6%	× 6.3 a 7.2%	1.44 6.4 a 7.2%	0.00 6.4 a 7.3%
AP_WRAP_SM	× 6.6%	× 7.1 a 7.2%	2.07 a 2.0 7.1 a 7.2%	0.00 7.2 a 7.3%

Bits inteiros I (F = 32)

TDLC300-100: N = 132, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14
AP_TRN	×	×	× 1.5 a 1.7%	× 1.7 a 1.8%	1.49 1.9 a 2.1%	0.03 2.0 a 2.2%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	2.49 1.9 a 2.3%	0.01 2.0 a 2.5%
AP_RND	×	×	× 1.6 a 1.9%	2.45 1.7 a 2.1%	0.07 2.0 a 2.3%	0.01 2.1 a 2.4%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	2.44 1.6 a 2.1%	0.07 1.9 a 2.3%	0.01 2.0 a 2.5%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	2.43 1.8 a 2.1%	0.07 2.1 a 2.3%	0.01 2.2 a 2.4%
AP_RND_INF	×	×	× 1.4 a 2.0%	2.44 1.6 a 2.1%	0.07 1.9 a 2.3%	0.01 2.0 a 2.5%
AP_RND_CONV	×	×	× 1.6 a 2.0%	2.46 1.8 a 2.1%	0.07 2.1 a 2.3%	0.01 2.2 a 2.5%

Bits fracionários F (I = 14)

TDLB100-400: N = 24, f_c = 3,5 GHz

Estouro	6	8	10
AP_SAT	× 5.5 a 6.6%	4.05 6.0 a 7.2%	0.00 6.1 a 7.2%
AP_SAT_ZERO	× 6.6%	4.05 7.1 a 7.2%	0.00 7.1 a 7.2%
AP_SAT_SYM	× 6.5 a 6.6%	4.05 7.1 a 7.2%	0.00 7.1 a 7.2%
AP_WRAP	× 5.8 a 6.6%	3.89 a 3.9 6.3 a 7.2%	0.00 6.4 a 7.2%
AP_WRAP_SM	× 6.6%	4.09 7.1 a 7.2%	0.00 7.1 a 7.2%

Bits inteiros I (F = 32)

TDLB100-400: N = 24, f_c = 3,5 GHz

Quantização	4	6	8	10	12	14	16
AP_TRN	×	×	× 1.5 a 1.7%	4.30 1.7 a 1.8%	0.27 1.9 a 2.1%	0.02 2.0 a 2.2%	0.00 3.1 a 3.7%
AP_TRN_ZERO	×	×	× 1.4 a 2.0%	× 1.6 a 2.1%	2.37 1.9 a 2.3%	0.09 2.0 a 2.5%	0.01 3.2 a 3.7%
AP_RND	×	×	× 1.6 a 1.9%	1.04 1.7 a 2.1%	0.05 2.0 a 2.3%	0.00 2.1 a 2.4%	0.00 3.5 a 3.7%
AP_RND_ZERO	×	×	× 1.4 a 2.0%	1.03 1.6 a 2.1%	0.05 1.9 a 2.3%	0.00 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_MIN_INF	×	×	× 1.6 a 1.9%	1.04 1.8 a 2.1%	0.05 2.1 a 2.3%	0.00 2.2 a 2.4%	0.00 3.7%
AP_RND_INF	×	×	× 1.4 a 2.0%	1.05 1.6 a 2.1%	0.05 1.9 a 2.3%	0.00 2.0 a 2.5%	0.00 3.2 a 3.7%
AP_RND_CONV	×	×	× 1.6 a 2.0%	1.04 1.8 a 2.1%	0.05 2.1 a 2.3%	0.00 2.2 a 2.5%	0.00 3.7%

Bits fracionários F (I = 14)